

An Approach for Analyzing and Predicting Traffic Accidents in the United States

(Identifying Risk Factors, Hotspots, and Predictive Modeling)

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Table of Contents

Introduction	4
Introduction/Background.....	4
Problem Statement.....	4
Problem Elaboration.....	5
Motivation.....	5
Project Scope.....	5
Literature Review	6
Relevant Research.....	6
Methodology	7
Dataset Description.....	7
Data Collection.....	7
Data Preprocessing.....	7
Feature Engineering.....	8
Data Visualizations.....	9
Data Modeling.....	13
Results & Analysis	17
Conclusion	19
Conclusion.....	19
Project Limitation.....	20
Future Research.....	20
References	21
Appendix	22
Appendix- A.....	22
Github Link.....	23

Glossary Of Terms

Accident Severity: The level of impact or damage caused by an accident, typically categorized into different levels (e.g., minor, moderate, severe) based on injuries and property damage.

Aggregate Statistics: Summarized data collected from multiple individual observations, used to identify overall patterns and trends.

Classification Model: A type of machine learning model that predicts a categorical label (e.g., predicting the severity level of an accident).

Cross-Validation: A statistical technique used to assess the generalization performance of a machine learning model by splitting the data into training and testing subsets multiple times.

EDA (Exploratory Data Analysis): The process of analyzing datasets to summarize their main characteristics, often using visual methods to uncover patterns, detect anomalies, and test hypotheses.

Feature Engineering: The process of selecting, modifying, or creating new input variables (features) to improve the performance of a machine learning model.

Geospatial Analysis: The use of location-based data to identify geographic patterns, clusters, or trends in accidents.

Hyperparameter Tuning: The process of optimizing the parameters that govern the training process of a machine learning model to improve its performance.

Machine Learning (ML): A field of computer science that uses algorithms to allow computers to learn patterns from data and make predictions or decisions without being explicitly programmed.

Model Evaluation Metrics: Quantitative measures (such as Accuracy, Precision, Recall, F1-Score) are used to assess the performance of a predictive model.

Predictive Modeling: The process of using historical data to build models that predict future outcomes.

Severity Prediction: The task of predicting how severe an accident will be based on input factors like weather, time of day, location, and road conditions.

Visualization: The graphical representation of data to help communicate insights, patterns, and findings effectively.

Introduction

Background

Road accidents represent one of the most significant challenges to public safety and transportation efficiency in the United States. Every year, millions of accidents lead to serious injuries, loss of life, and substantial economic costs, including medical expenses, property damage, and lost productivity. Despite improvements in vehicle technology, infrastructure, and traffic regulations, the rate of accidents remains alarmingly high, indicating the complexity and multifactorial nature of the problem. With the growing availability of large-scale traffic, environmental, and demographic data, there is a tremendous opportunity to leverage data-driven methods to analyze accident patterns, uncover hidden relationships among contributing factors, and predict accident severity more accurately. Advanced machine learning techniques can capture complex, non-linear interactions between variables, offering predictive insights that traditional methods often miss.

Problem Statement

Despite numerous initiatives to improve road safety, traffic accidents continue to occur at an alarming rate across the United States. Traditional analysis methods often fail to capture complex, non-linear relationships between environmental factors, location-specific attributes, and accident severity. There is a pressing need for predictive systems that can analyze historical accident data and forecast accident severity with high accuracy to assist policymakers, transportation departments, and emergency response teams.

Problem Elaboration

The complexity of accident causation stems from multiple interdependent factors such as weather conditions, time of day, road features, and traffic patterns. Traditional statistical models offer limited predictive power in such multifactorial scenarios. Machine learning, however, presents a robust solution by modeling complex relationships without heavy assumptions about data distribution. This project focuses on analyzing a large, multi-year dataset of US traffic accidents to predict severity outcomes and understand key factors influencing accidents through comprehensive exploratory data analysis (EDA), feature engineering, and model evaluation.

Motivation

Inspired by the increasing role of data science in solving real-world societal challenges and the growth of machine learning applications in public safety, this capstone project aims to analyze and predict the severity of traffic accidents across the United States. The benefits of this approach include the ability to assist policymakers, transportation authorities, and emergency services in making data-driven decisions, improving resource allocation, and ultimately enhancing road safety outcomes.

Project Scope

Traffic accident data across the United States is collected from public sources to analyze and predict accident severity. The dataset includes factors such as location, weather, time, and road conditions. The purpose is to develop machine learning models capable of accurately predicting accident severity and to identify key factors influencing accident outcomes through exploratory data analysis and visualization techniques.

Literature Review

Research studies have explored the use of data analytics and machine learning techniques to understand and predict traffic accident severity. In the study “Comprehensive Analysis of Road Traffic Accidents in the United States,” the authors used big data methods to identify critical patterns related to accident frequency, weather impacts, and severity outcomes. Their work emphasized the significance of leveraging large datasets and visualization tools for uncovering meaningful insights into accident trends across different regions.

Another study, “Modern data sources and Techniques for Analysis and Forecast of Road Accidents,” implemented Support Vector Machines (SVM) and Random Forest classifiers to predict accident severity using variables such as weather, road conditions, and temporal factors. However, the study highlighted limitations due to the absence of hyperparameter tuning and detailed feature selection, which affected model optimization and predictive performance.

Additionally, research “Road Traffic Accidental Injuries and Deaths: A Neglected Global Health Issue,” focused on analyzing demographic influences on road accident outcomes using logistic regression techniques. While informative for understanding certain population-based trends, the study was limited by its use of traditional statistical methods and lacked the integration of more complex machine learning models or spatial analysis.

Building upon insights from these studies, this project advances prior research by applying modern machine learning algorithms, performing hyperparameter tuning, conducting comprehensive exploratory data analysis (EDA), and developing interactive dashboards to enhance the understanding of accident severity factors.

Methodology

Dataset Description

The dataset used in this project is sourced from a publicly available collection of U.S. traffic accident records, covering incidents from 2016 to 2023. It includes over 7 million accident entries across 49 states. Each record captures detailed information about an individual accident event, including attributes such as timestamp, location (latitude, longitude, city, state), weather conditions, road conditions, visibility, traffic signal presence, and accident severity. The dataset provides a comprehensive view of environmental, temporal, and infrastructural factors influencing accidents, making it ideal for predictive modeling and exploratory data analysis.

Data Collection

The dataset was collected from open data platforms, particularly Kaggle and official traffic incident databases that aggregate reports from departments of transportation, law enforcement agencies, and traffic sensors. Data acquisition focused on ensuring diversity across geographies and accident types to build a representative sample. The raw data was downloaded in CSV format and loaded into a cloud-based environment for processing and analysis. No additional web scraping or real-time data integration was performed, maintaining consistency with static historical datasets.

Data Preprocessing

Before modeling, the dataset underwent essential preprocessing steps to ensure quality and consistency. These preprocessing steps prepared a clean and structured dataset suitable for feature engineering and model training.

Handling Missing Values: Features with excessive missing data were either imputed or removed based on relevance.

Categorical Encoding: Categorical variables (e.g., weather condition, city, state) were encoded using label encoding and one-hot encoding as appropriate.

Outlier Treatment: Extreme values in continuous variables such as temperature, visibility, and wind speed were inspected, but extreme removal was selectively avoided to maintain natural variance.

Feature Selection: Based on domain knowledge and correlation analysis, important features were selected for modeling, including weather conditions, road types, distance, and time of occurrence.

Feature Engineering

Feature engineering was a critical step in preparing the accident dataset for predictive modeling. Several new features were created, and existing ones were transformed to enhance model performance and capture important patterns in the data. Several new features were engineered to improve the model's predictive ability:

Time Features: Extracted year, month, day, hour, and weekday from accident start times.

Created indicators for rush hour and weekend accidents.

Spatial Features: Applied KMeans clustering to accident locations and created `distance_to_nearest_hotspot` to measure proximity to accident-prone areas.

Road Complexity: Engineered `Is_Complex_Road` based on the presence of junctions, railways, and crossings.

Traffic and Lighting Conditions: Converted fields like Traffic_Signal and Sunrise_Sunset into binary formats for modeling.

Accident Duration: Calculated the duration of accidents and applied a log transformation for normalization.

Weather Encoding: Filled missing weather conditions and applied one-hot encoding to weather and wind direction features.

Feature Reduction: Removed non-predictive fields such as City, County, Street, Zipcode, State, and Country.

Data Visualization

Visualizations were a key component of this project to better understand the distribution of accidents, identify high-risk patterns, and support the interpretation of model results. Through graphical analysis, complex and large volumes of data were translated into intuitive visual formats, enabling easier discovery of underlying trends and anomalies. Visual exploration revealed important insights related to the time of occurrence, geographical distribution, weather conditions, and severity levels of accidents.

By using time-based, spatial, and categorical visualizations, we were able to highlight critical periods of increased accident frequency, recognize accident-prone locations, and assess the influence of environmental factors such as visibility, precipitation, and road surface conditions. Furthermore, visualizations of model outputs, such as feature importance plots and confusion matrices, provided deeper understanding into the behavior and effectiveness of machine learning models. Overall, visual analysis not only enhanced exploratory data analysis but also played a vital role in communicating the project findings to both technical and non-technical audiences.

A variety of graphs were created to visually represent patterns in the accident data and model results. Time-based graphs (such as yearly, monthly, and hourly distributions) helped identify accident trends over time, while spatial visualizations like heatmaps highlighted high-risk locations. Severity distribution graphs provided insights into the impact levels of accidents. Together, these visualizations made complex data easier to interpret and helped support data-driven conclusions.

The following visualizations summarize key accident patterns and model insights discovered throughout the analysis

Fig.1 : Severity Distribution

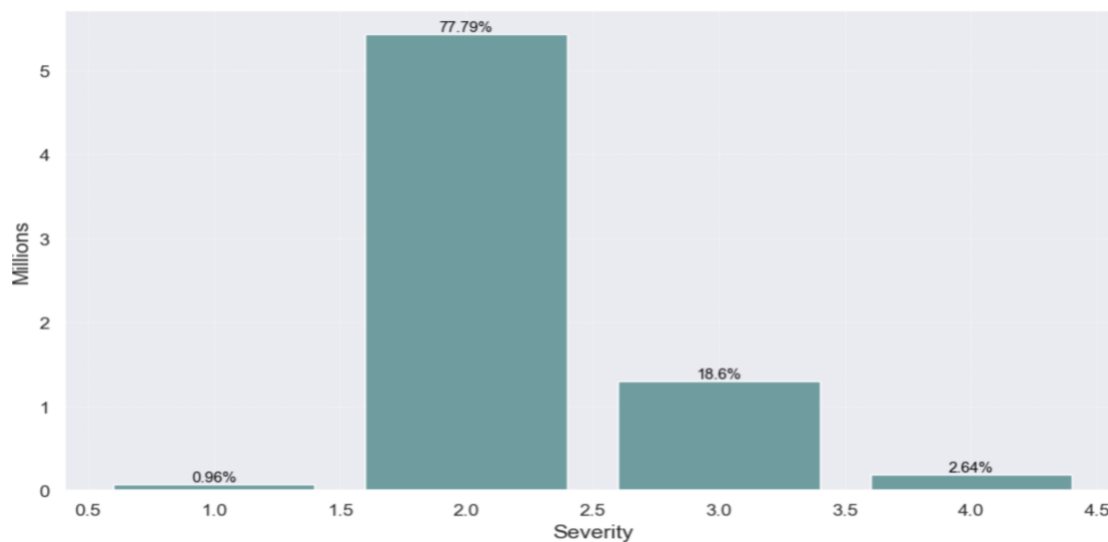


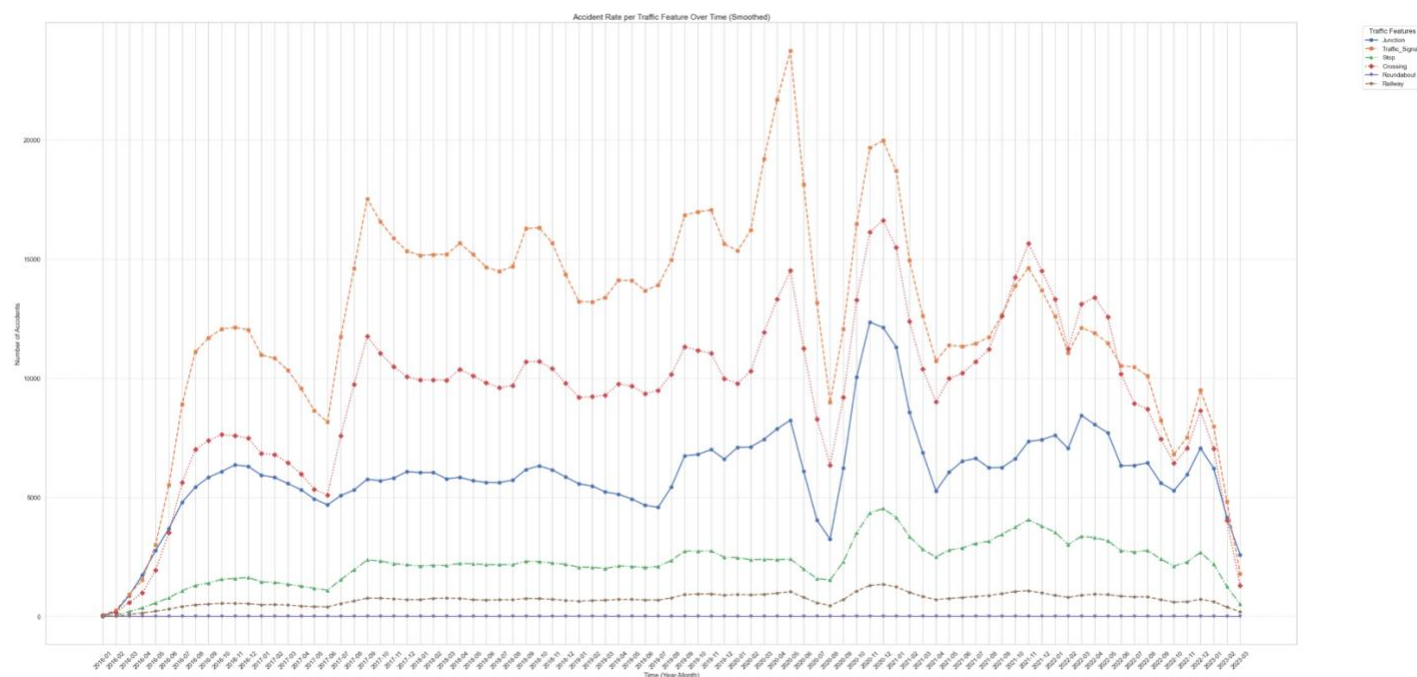
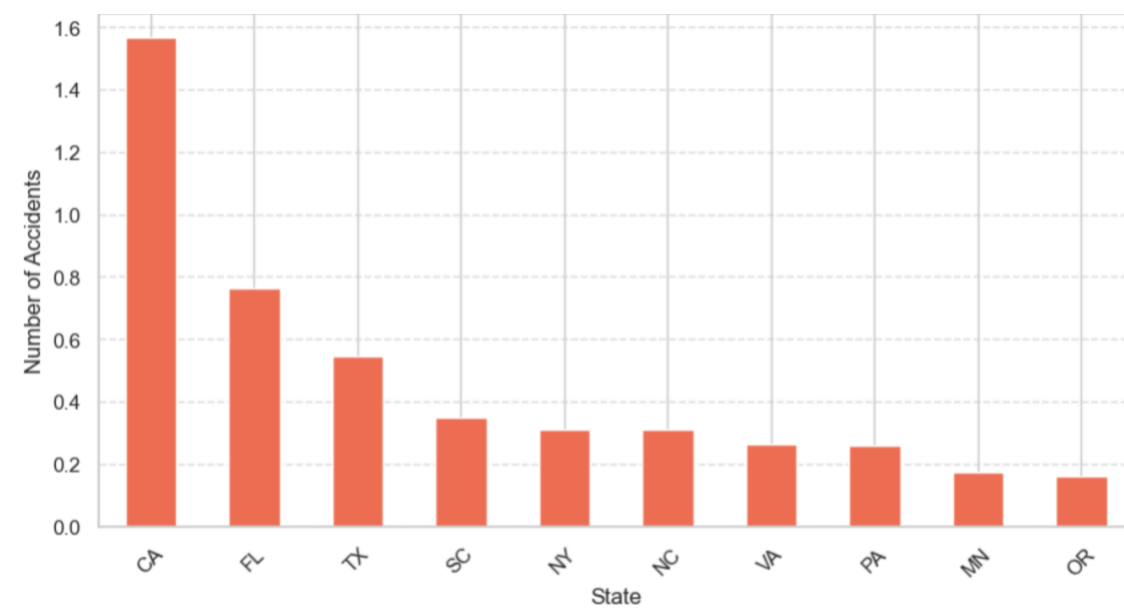
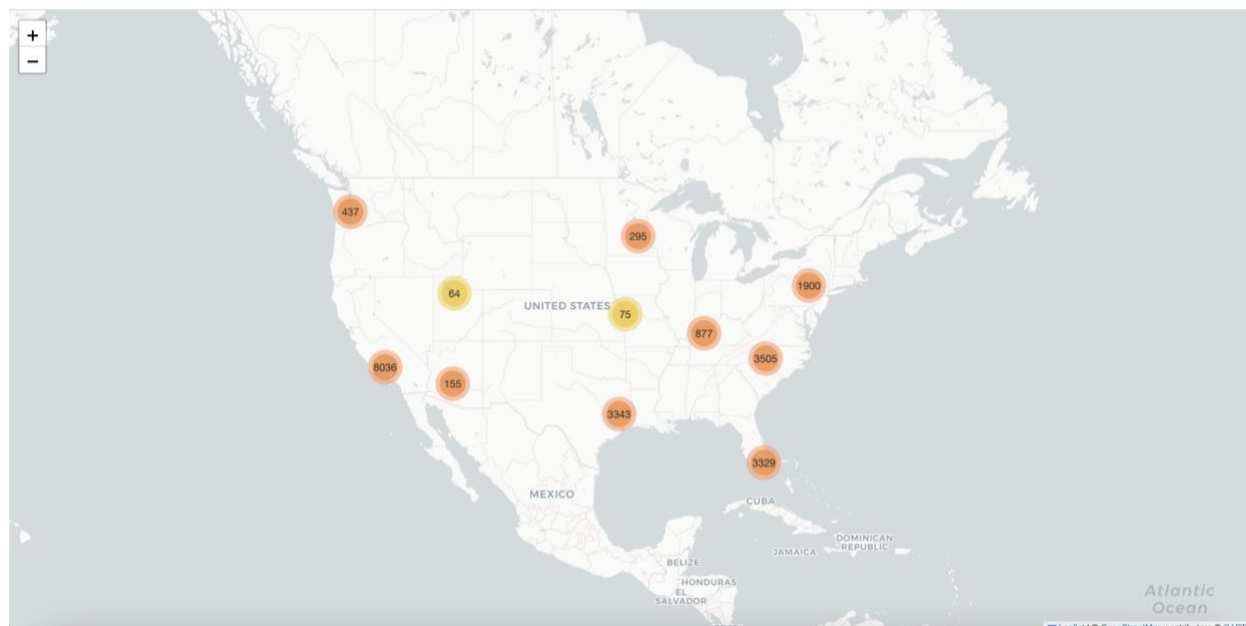
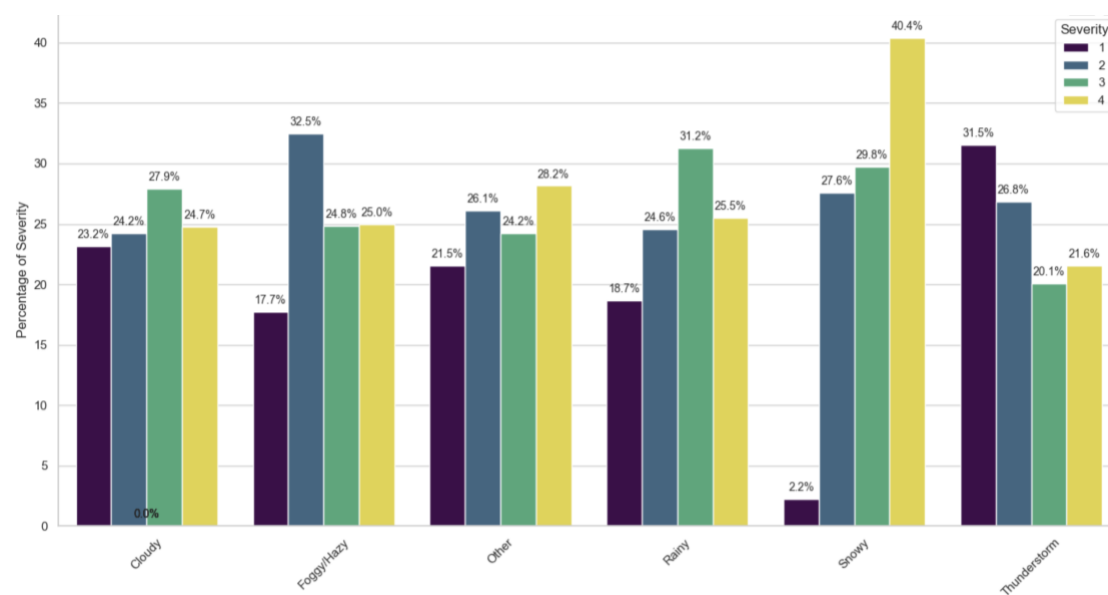
Fig.2: Yearly Accident Trend**Fig.3: Top 10 States with Accidents**

Fig.4: High-risk Accident Hotspot Clusters**Fig.5: Accident Severity by Weather Condition**

Data Modeling

The goal of data modeling in this project was to predict the severity of traffic accidents using historical data. Several machine learning models were developed, trained, and evaluated to identify the best-performing approaches for severity classification. The models used included:

Logistic Regression — as a baseline model to establish initial performance benchmarks.

Random Forest Classifier — to capture non-linear relationships and feature interactions.

XGBoost Classifier — for optimized boosting performance with high prediction accuracy.

CatBoost Classifier — to efficiently handle categorical variables without extensive preprocessing.

LightGBM Classifier — for faster training with large datasets and superior scalability.

Each model was trained on the preprocessed dataset and evaluated using cross-validation techniques to ensure robust generalization. Key performance metrics such as Accuracy, Precision, Recall, and F1-Score were used to compare models.

Fig.6: Model performance comparison

Model	Accuracy Mean	Precision (Weighted Mean)	Recall (Weighted Mean)	F1-Score (Weighted Mean)
Logistic Regression	0.77	0.60	0.77	0.68
Random Forest Classifier	0.79	0.77	0.79	0.77
XGBoost Classifier	0.85	0.84	0.85	0.83
CatBoost Classifier	0.82	0.80	0.81	0.80
LightGBM Classifier	0.81	0.79	0.81	0.78

Hyperparameter tuning was performed specifically for the two best-performing models XGBoost Classifier and CatBoost Classifier using RandomizedSearchCV. A range of important

hyperparameters, such as the number of estimators, learning rate, maximum depth, and subsample ratio, were systematically explored to optimize model performance.

Through tuning, both models achieved improved predictive accuracy, better generalization on unseen data, and enhanced stability. Hyperparameter optimization also helped address the class imbalance within accident severity levels by improving precision and recall across minority classes. The tuned versions of XGBoost Classifier and CatBoost Classifier demonstrated more robust and consistent results compared to their default settings, strengthening the reliability of severity predictions made by the models.

Fig.7: Performance Comparison before and after tuning.

Metric	XGBoost (Before Tuning)	XGBoost (After Tuning)	CatBoost (Before Tuning)	CatBoost (After Tuning)
Accuracy	0.85	0.89	0.81	0.87
Precision (Weighted Avg)	0.84	0.89	0.80	0.86
Recall (Weighted Avg)	0.85	0.89	0.81	0.87
F1-Score (Weighted Avg)	0.83	0.89	0.80	0.86

Feature importance refers to techniques that assign a score to input features based on how useful they are at predicting a target variable. In machine learning models, especially tree-based models like XGBoost Classifier, feature importance helps identify which variables most strongly influence the model's predictions.

Understanding feature importance provides several benefits:

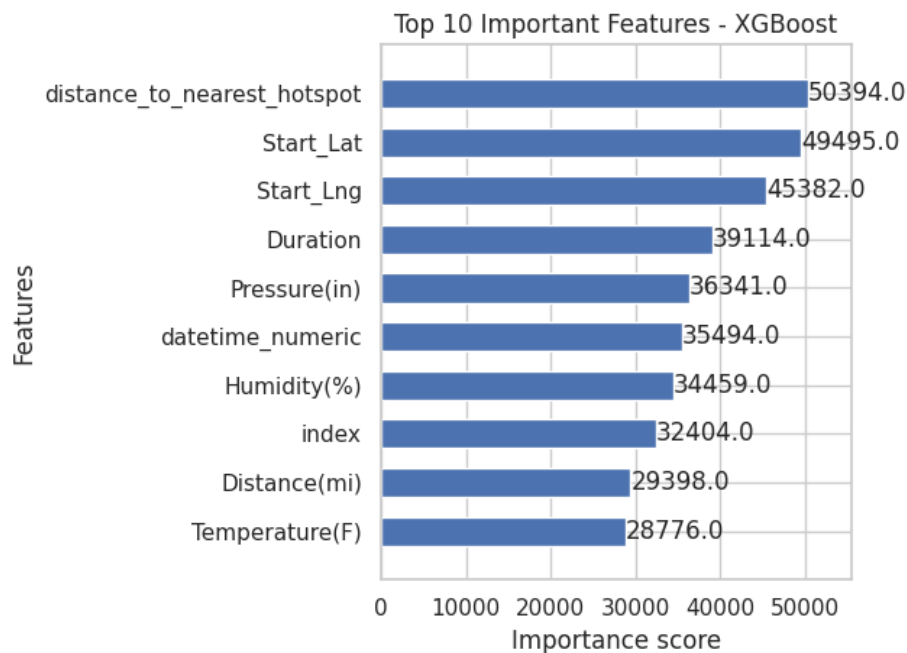
It improves model interpretability by explaining which features drive predictions.

It helps in feature selection, allowing future models to focus on the most relevant inputs.

It offers actionable insights into key factors associated with higher accident severity, which can inform real-world road safety strategies.

For this project, feature importance was calculated from the best-performing model, XGBoost Classifier.

Fig.8: Key features impacting severity prediction.



The most influential features for predicting accident severity include distance to nearest hotspot, latitude/longitude, and duration. Environmental factors like pressure, humidity, and temperature also play a key role, highlighting the importance of both location and weather conditions in model predictions.

Overall, the modeling process demonstrated that advanced ensemble models provided superior prediction performance compared to simpler models, and highlighted the key environmental and temporal factors contributing to accident severity.

Results and Analysis

The project successfully developed multiple machine learning models to predict the severity of traffic accidents. Initial modeling results showed that ensemble methods, particularly XGBoost Classifier and CatBoost Classifier, outperformed traditional models like Logistic Regression and Random Forest in terms of accuracy, precision, recall, and F1-score.

Among all models, XGBoost Classifier achieved the highest performance before tuning, with an accuracy of 85%, followed by CatBoost Classifier at 82%.

After applying hyperparameter tuning using RandomizedSearchCV, the performance of both models significantly improved:

XGBoost Classifier Accuracy improved from 0.85 to 0.89.

CatBoost Classifier Accuracy improved from 0.81 to 0.87.

Other evaluation metrics, including Precision, Recall, and F1-Score, also saw notable increases for both models after tuning, indicating enhanced stability and better handling of class imbalance in severity classes.

The feature importance analysis from the XGBoost Classifier model revealed that key factors influencing accident severity included:

Distance to nearest accident hotspot

Geographical coordinates (Start_Lat, Start_Lng)

Weather factors such as Pressure, Humidity, and Temperature.

Additionally, visual analysis highlighted important accident patterns:

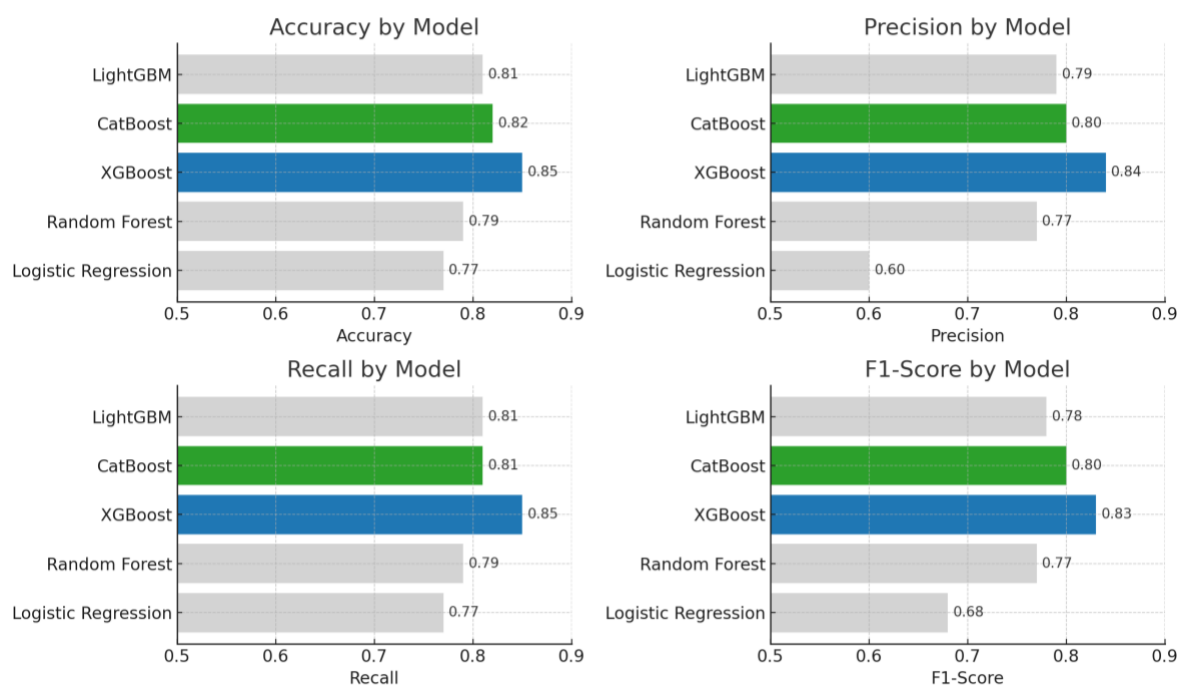
Accidents were most frequent during morning and evening rush hours.

Higher accident counts were observed in urban areas and during adverse weather conditions like rain and fog.

Severity distribution showed that most accidents were low to moderate severity, with fewer but significant high-severity cases.

Overall, the results demonstrate the effectiveness of advanced ensemble models in predicting accident severity, and the visualizations provided critical insights into temporal, environmental, and geographic accident patterns.

Fig.9: Model Comparison



Conclusion

This project successfully demonstrated the application of machine learning techniques to analyze and predict the severity of traffic accidents across the United States. By conducting detailed exploratory data analysis, critical patterns and correlations between accident factors and severity outcomes were uncovered. Multiple machine learning models, including Logistic Regression, Random Forest, XGBoost Classifier, CatBoost Classifier, and LightGBM Classifier, were developed and rigorously evaluated. Among these, ensemble models such as XGBoost Classifier and CatBoost Classifier consistently delivered the highest predictive performance, highlighting the strength of advanced ensemble learning methods in handling complex, real-world data.

Hyperparameter tuning through RandomizedSearchCV further enhanced the models' performance by improving accuracy, precision, recall, and F1-scores, thereby demonstrating the significant impact of optimization strategies on model effectiveness.

Feature importance analysis revealed that variables such as distance to accident hotspots, geographic coordinates, and environmental factors like pressure, humidity, and temperature played critical roles in influencing accident severity. These insights not only validated the models' interpretability but also provided valuable information for practical road safety interventions.

Overall, the findings from this project contribute to a deeper understanding of accident dynamics, demonstrate the feasibility of predictive modeling for public safety applications, and reinforce the importance of data-driven approaches in designing strategies to reduce the severity and impact of traffic accidents in the future.

Project Limitations

Despite achieving promising results, the project faced several limitations:

Feature Availability: The dataset primarily included environmental and locational attributes. Important variables like driver behavior (e.g., distraction, fatigue), vehicle type, and real-time traffic congestion were not available, which could have further improved model accuracy.

Resource Constraints: Due to the large dataset size, hyperparameter tuning and model training were computationally intensive and time-consuming, limiting the ability to explore even deeper model architectures or more extensive grid searches.

Future Research

There are opportunities to expand this work in the future:

Incorporating Behavioral and Vehicle Data: Future studies could integrate datasets that include driver conditions, vehicle types, and in-vehicle sensor data for more comprehensive severity prediction.

Real-time Accident Prediction Systems: Extending the models to work with real-time data feeds (such as live traffic, weather, and sensor data) can help develop early warning systems for accident prevention.

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Appendix

Appendix-A

Supplementary Data Visualizations

To provide a more comprehensive understanding of traffic accident patterns and environmental influences, additional exploratory visualizations were developed.

These graphs support the main findings and offer deeper insights into trends, temporal patterns, geographical distributions, and severity factors.

Fig.10: Monthly Distribution of Accidents

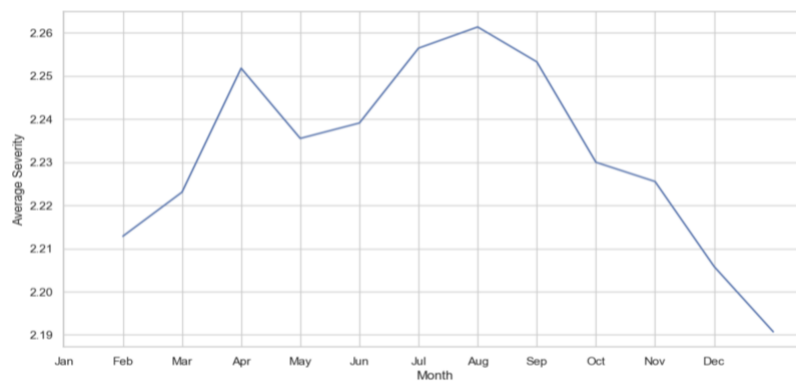


Fig.11: Accident Density with Weather Condition



Fig.12: Geographic Distribution of Accident Severity

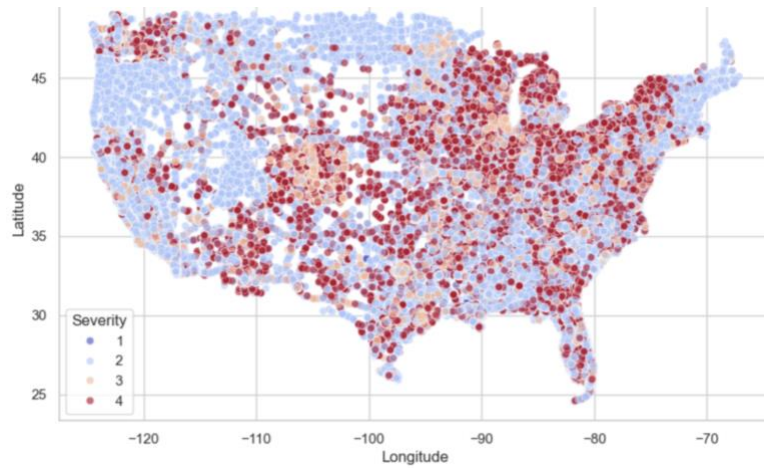
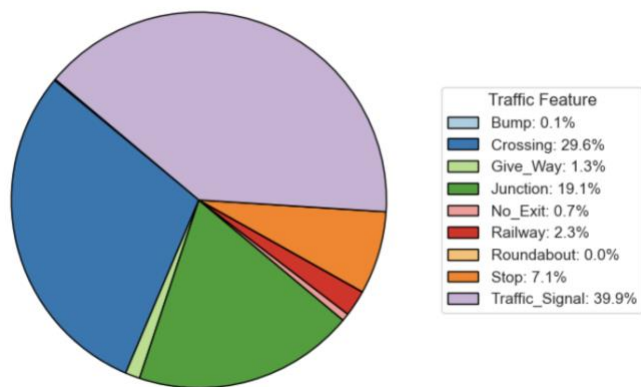


Fig.13: Proportion of Accidents near Traffic Features



Github link: <https://github.com/JagadeeshKalyanapu/An-Approach-for-Analyzing-and-Predicting-Traffic-Accidents-in-the-United-States>